UNDERGRADUATE RESEARCH OPPORTUNITIES PROGRAMME
(UROP)

PROJECT REPORT

Diagnostics & Repair of Boston Dynamics Spot

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1. Introduction

The Boston Dynamics *Spot* quadruped was received non-operational, and this report documents the hardware-level work carried out to return it toward service.

Two independent faults were present on arrival. First, the power subsystem was dead: both of Spot's battery modules refused to charge. Second, once external power was restored, one leg actuator, the left hind abduction/adduction joint (hereafter *hl.hx*), behaved abnormally (Figure 1), with the joint failing to reach commanded positions while the other three legs moved normally.

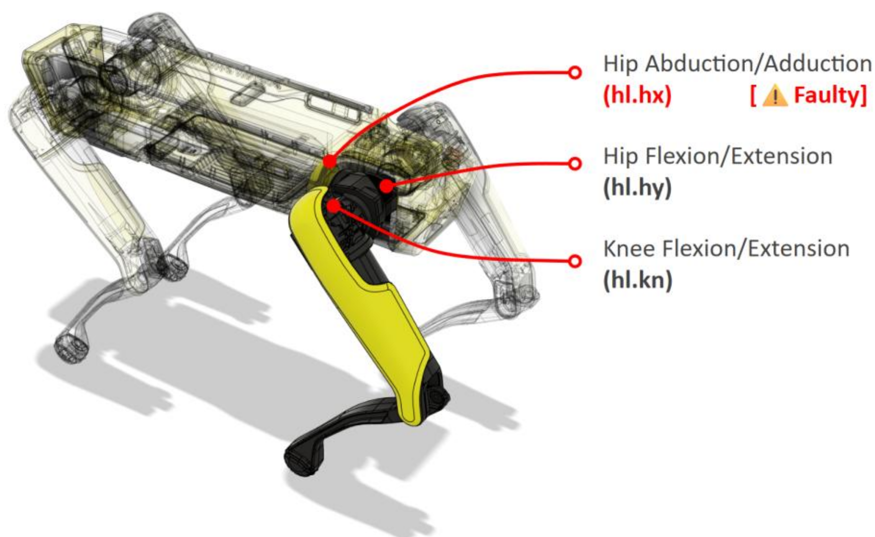

Figure 1: Spot joint names and identified faulty joint, *hl.hx*.

The first objective was to restore the power subsystem by diagnosing the charge fault. Should the cells prove unrecoverable, each pack will be rebuilt on its original battery management system (BMS) in the original-equipment-manufacturer (OEM) form factor. The second objective was to narrow the actuator fault to a specific component through a sequence of component swaps.

The scope of this report focuses on the hardware layer and the low-level electrical layer. It describes physical tests, facts observed, and a low-level analysis of the electrical subsystem. The complementary *diagnostic interpretation* is reported in the companion report by a groupmate working on the same robot platform.

The working approach throughout was bench-based and comparative. Because Spot's two hind legs are identical, the healthy hind leg served as a reference against which the faulty one is compared against component wise.

2. Background

2.1 Battery module architecture

The Spot enterprise model uses a battery pack distinct from that of the base Spot, identifiable by its finned aluminum backplate. The battery module is a sealed assembly whose perimeter is held closed by Room Temperature Vulcanizing (RTV) silicone compound. The internal architecture was first established from a published third-party teardown of the Spot platform[1] and then confirmed directly during the project's own disassembly. Table 1 summarizes the construction as found.

Nominal energy	~500 Wh
Rated runtime	~1.5 h
Cell	Samsung INR18650-30Q (3000 mAh, 15 A cont. / 30 A peak)
Cell count & topology	56 cells in 14s4p, built as two 7s4p sub-packs (A & B)
BMS	Single PCB, STM32F1xx MCU, CAN bus to the robot
Measurement pads	28 pads: positives along the long PCB edges, common negative down the center
Thermal sensing	One flexible PCB per sub-pack (A/B), each carrying five SMD negative-temperature-coefficient (NTC) thermistors
Thermal / safety	Latent Heat Solutions phase-change material (PCM) casing around the cells; fish-paper terminal insulation; foam seating

Table 1: Battery module architecture

The irregular OEM cell layout is described in Section 3.4. Thermal management is provided by a phase-change material casing custom made by Latent Heat Solutions (LHS)[2], which absorbs heat through its latent heat of fusion and buffers thermal-runaway energy

2.2 Motion system and joint nomenclature

Each of Spot's legs is a self-contained 4.2kg drive module built around 3 identical motors, with 2 rotary joints (X and Y axis) driven through 50:1 harmonic drives and the knee joint actuated by a linear ball-screw system that turns rotational motion from the knee (Z-axis) drive motor into push-pull action to swing the lower leg[3]. Boston Dynamics joint nomenclature identifies each leg by a two-letter code combining its longitudinal position (front or hind) with its lateral position (left or right), giving fl, fr, hl, and hr. Individual joints are then referenced by appending a suffix to the leg code: hx for the hip X (abduction/adduction) joint, hy for the hip Y (flexion/extension) joint, and kn for the knee[4]. The front left leg, for example, is described by *fl.hx*, fl.hy, and fl.kn.

3. Power-system rebuild

3.1 Fault diagnosis and the decision to rebuild

The power subsystem was the first blocker: both battery modules refused to charge. When attempting to charge, the charge indicator flashed red briefly before turning off, rather than the steady green blink that indicates normal charging (Figure 2).

A module was unscrewed and opened along its adhesive seam. The initial cell-group voltages averaged only about **90 mV**, far below the 2.5 V minimum safe voltage for an 18650 cell, indicating severe over-discharge.

A staged recovery was attempted: each accessible group was bench-charged slowly at 2.0 V / 1.5 A across the BMS tabs, and once groups reached roughly 2.0 V the limits were raised to 16 V / 5 A across up to five groups at a time (Figure 3). The pack rose to about 14.5 V on the supply (P1-GND measured 13.3 V combined), demonstrating that it would take charge.

Figure 2: Charging indicator turns off after blinking red, indicating a charge fault.

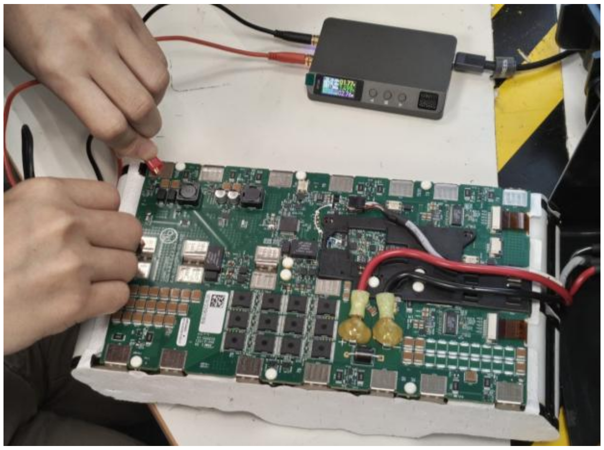

Figure 3: Trickle charging of the individual cell groups.

After each group is manually charged to about 3.0V, a green LED beside the STM32 MCU blinked, indicating the cells have enough charge to power the control circuitry. The module was tested with the charger again, but the charge indicator remained a static green, reporting a *"full"* charge on cells that were obviously discharged (Figure 4). The Spot manual attributes a false "full" to highly imbalanced cells and suggests leaving the pack plugged in to auto-rebalance, which had no effect. After idling, the cells were warm to the touch and individual cell-group voltages collapsed from about 1.6 V to 0.6 V. The combination of warmth and rapid voltage collapse is consistent with resistive self-discharge through dead cells now behaving as internal short circuits[5]; the cells were therefore judged degraded beyond recovery. Hence, a replacement of all cells was mandatory.

Figure 4: Battery state of charge (SoC) reports no charge, but charger reports full charge (static green indicator).

3.2 Teardown

Having resolved to rebuild, the module was fully disassembled, both to strip out the dead cells and to document the OEM construction so the replacement pack could match it. The power cables were freed by cutting away a large securing glue blob piece by piece, then unscrewing the leads and unplugging the CAN cable; a plastic insulator covering three of the BMS pad sets was removed (Figure 5). The nickel strips are spot-welded to the BMS pads so they were pried off (Figure 6). The two 7s4p sub-packs were then separated by removing a long plastic snap-connector holding the 2 packs together (Figure 7). Two flexible thermistor strips used to monitor pack temperature were unplugged and labelled A and B to preserve their original sides (Figure 8).

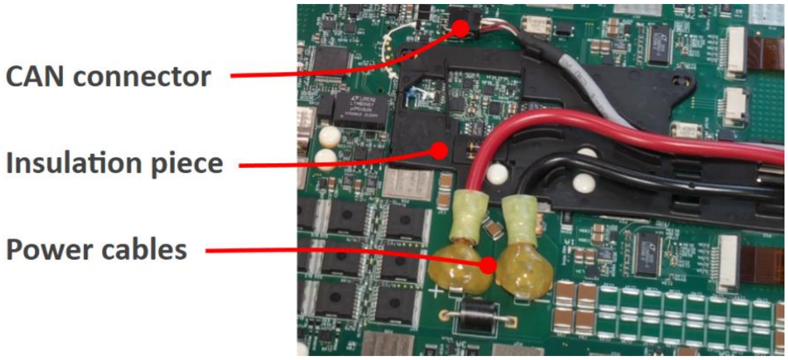

Figure 5: BMS components detached.

Figure 6: Nickel tabs pried off BMS pads.

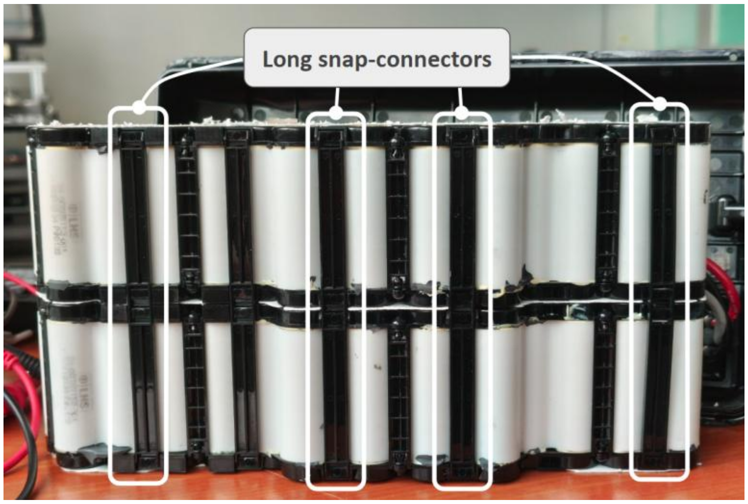

Figure 7: Snap connectors holding packs together.

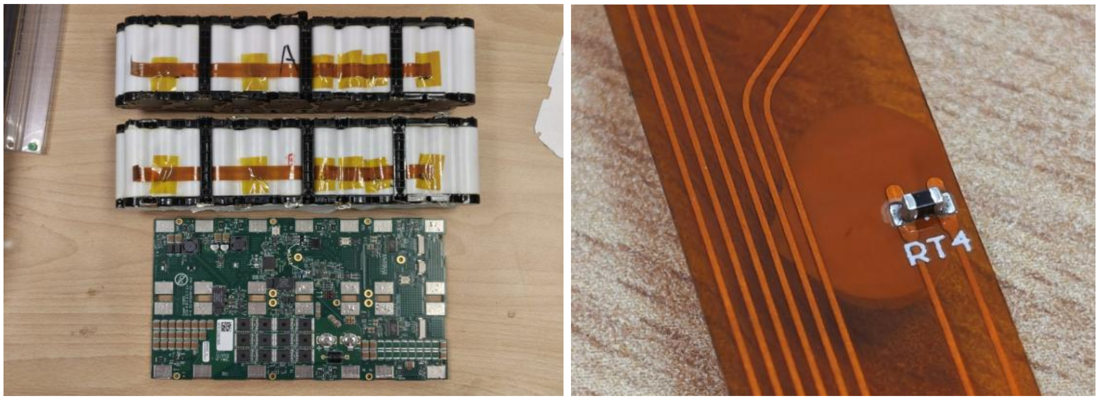

Figure 8: Fully separated module, close-up of single NTC thermistor on flex PCB.

The OEM cell arrangement was dimensioned (Appendix I). The BMS pads layout is also analyzed; the pads are numbered in a weird order, with the groups connected in series from P1 to P28 in order (Figure 9). The full pack voltage is measured across P1 (positive) and P28 (negative).

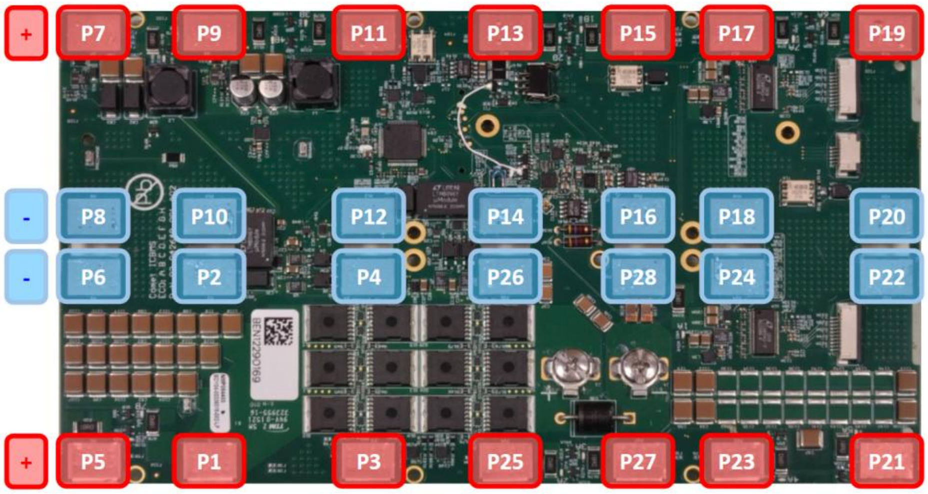

Figure 9: BMS pad layout, negative terminal along center, positive along long edge.

3.3 Cell sourcing

The replacement cell was fixed by the OEM specification: the Samsung INR18650-30Q identified from the Spot Battery Safety Data Sheet[6], of which 56 were required per module. The cheapest source was from Falcon PEV[7].

3.4 Pack design V1

Two design questions had to be settled before assembly: inter-cell nickel strip sizing, and how to hold the irregular cell lattice rigidly.

The conductor sizing followed from an estimate of Spot's power draw. The OEM pack design uses 5x0.15mm nickel strips between cells and 12x0.15mm for combined group output, leaving plenty of margin (Appendix C). We decided to use 8mm wide strips between cells as it was a more common size and gives even more current headroom.

The OEM cell layout is irregular: the cells are unevenly spaced and are not even collinear in the middle row. This layout was replicated in CAD (Appendix I) from direct measurement of the original pack and 3D-printed as a 2-piece cell spacer in **PC-CF** (polycarbonate + carbon-fiber), chosen for its high operating-temperature limit and dimensional stability (Figure 10). An off-the-shelf phase-change material equivalent to the OEM latent-heat filler [2] was researched but no solutions easy to work with was found, so it was omitted from this rebuild, on the basis that joule-heating due to internal impedance of the lithium cells can be passively dissipated even under peak load (Appendix B).

Figure 10: 3D-printed PC-CF cell spacers replicating OEM cell positions.

3.5 Assembly and spot-welding

The cells were seated in the spacers and their terminals joined with nickel strips (Figure 11). Only 10×0.15 mm and 12x0.2mm strip was available, so it was trimmed to about 8 mm before use. The 12 mm BMS PCB tabs were thicker and tended to pop off since they were stacked on up to 2 layers of 8mm strips (Figure 12), hence some were soldered rather than welded.

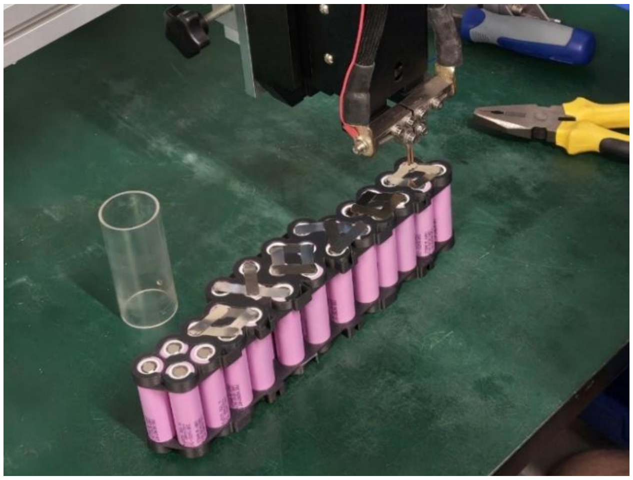

Figure 11: Spot welding of nickel strips.

Figure 12: Unsecure welds due to strip stacking.

The pack was then spot welded to the BMS, ground ends first and positive ends second, in no specific order. The module was finally closed with its original foam pieces reinserted to keep the cells seated (Figure 13).

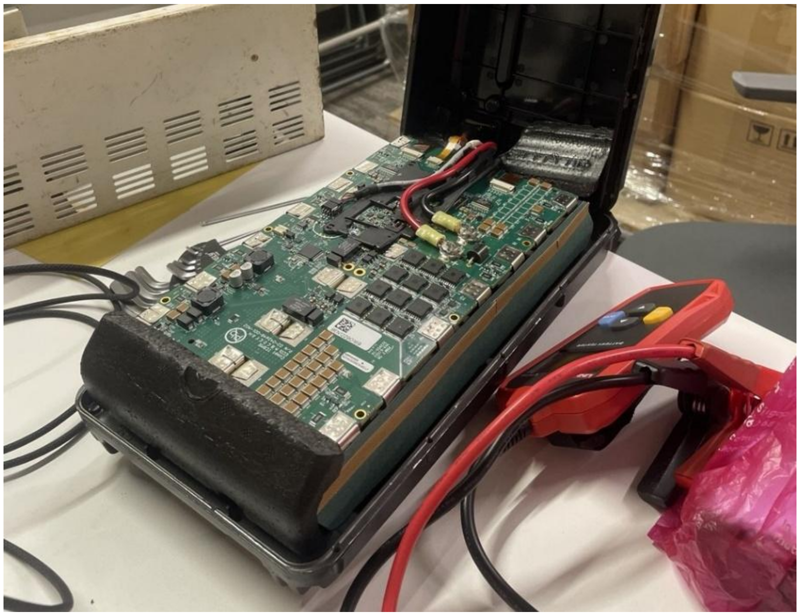

Figure 13: Reconstructed cell pack inserted back into the battery module shell.

4. Pack testing and fault-finding

4.1 First charge test and cell-group imbalance

Before the first charge test, the state-of-charge (SoC) button showed a single blinking green bar indicating a low initial charge. The charger flashed green (charging) but, after about twenty minutes, went static green ("full") while the SoC only showed one bar (Figure 14). Red indicator

LEDs on the BMS flashed when the charging halted, indicating the BMS was trying to balance the cell voltages, pausing the charge.

This observation was unexpected as half the cells were randomly tested before installation, and the initial charge was within 3.6 ± 0.05V. Hence, the voltage imbalance issue was attributed to the charging process.

Figure 14: Static green light on charger when pack is obviously discharged.

While the pack was unable to charge beyond a single SoC bar (~20%), it was able to power up Spot when inserted. We successfully connected to Spot via access point and gained access to the admin console, where the battery balance index read **0.213** (Figure 15), well above the 0.1 threshold for a healthy pack. Boston Dynamics suggests leaving the pack to auto-balance or connecting the pack to the charger overnight for active cell rebalancing[8]. The latter, however, requires battery firmware later than V45[14], whereas the pack reported V33 in the admin console[8]. The pack was left unconnected overnight in an attempt for the cells to passively rebalance.

Battery Information

Firmware Version	V33
State-of-Charge	14%
State	UNBALANCED
Voltage	48.314
Current	3.874
Log Capacity Used	2%
Recharge Cycle Count	90
CAN Node Serial Number	a2-21807-000c

Battery State

Cell Voltage Differential	0.213

Figure 15: Highly imbalanced cell voltage differential displayed in admin console.

After charging until the charger again reported “full”, each group’s voltage was measured and tabulated (Figure 16). Inspection reveals the groups with abnormal readings were caused by bad welds on the positive side: on groups 2 and 8 the positive weld had popped off, leaving only two of the four parallel cells connected; on group 1 a defective weld loosely joined the nickel strips, producing a high contact resistance (Figure 17). A resistance comparison across sound and poor joints (Figure 18) confirmed that an inconsistent weld raises the trace resistance substantially.

This logically explains the cell imbalance issue: groups with defective welds will charge faster since there are less batteries connected in parallel, causing the imbalance.

Group No.	BMS Pad no.	Voltage	
		Test 1	Test 2
1	1-2	3.64	3.82
2	3-4	3.75	3.96
3	5-6	3.64	3.75
4	7-8	3.64	3.75
5	9-10	3.64	3.75
6	11-12	3.64	3.75
7	13-14	3.64	3.75
8	15-16	3.99	3.75
9	17-18	3.64	3.75
10	19-20	3.64	3.75
11	21-22	3.64	3.75
12	23-24	3.64	3.75
13	25-26	3.64	3.75
14	27-28	3.64	3.75

Figure 16: Per-group voltage sweep, abnormal group voltages highlighted in red.

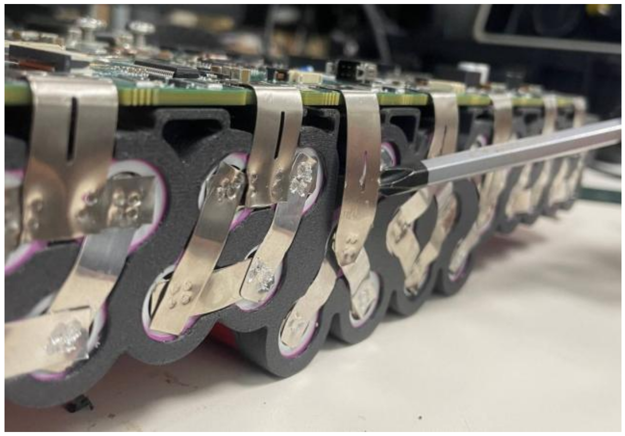

Figure 17: A defective / popped weld joint responsible for the group imbalance.

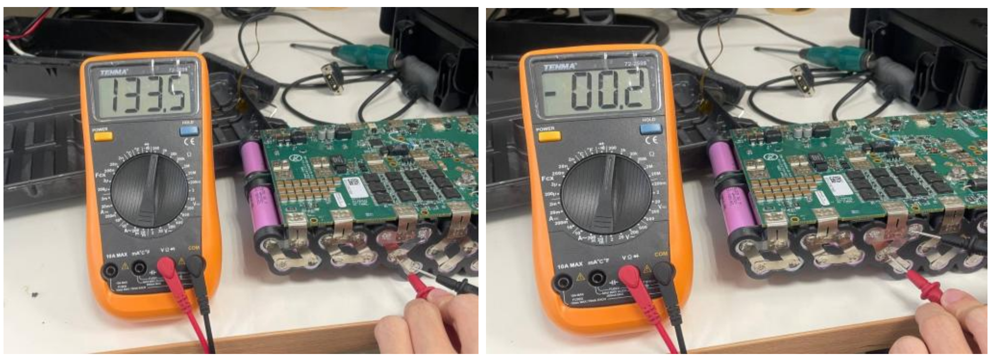

Figure 18: Substantially higher resistance across a poor weld (left) versus a good one (right).

4.2 Hard lockout and the broken negative-terminal weld

The weld problem then escalated from imbalance to a hard lockout. After the pack was left overnight, Spot reported a **battery fault** and refused to power on. Another voltage sweep (Figure 19) showed group 1 reading an impossible **−0.5 V** while every other group sat at a healthy 3.71-3.72 V. The group only read 3.71 V when the multimeter probe was reached down and pressed directly onto a cell's negative terminal (Figure 20). That intermittent contact isolated the fault to a broken weld on the group's negative terminal: the group had effectively disconnected from the BMS, and the −0.5 V was a phantom reading, most plausibly the reverse bias of an internal protection diode.

Group No.	BMS Pad no.	Voltage		
		Test 1	Test 2	Test 3 (9/6/26)
1	1-2	3.64	3.82	-0.5
2	3-4	3.75	3.96	3.72
3	5-6	3.64	3.75	3.72
4	7-8	3.64	3.75	3.73
5	9-10	3.64	3.75	3.73
6	11-12	3.64	3.75	3.72
7	13-14	3.64	3.75	3.72
8	15-16	3.99	3.75	3.71
9	17-18	3.64	3.75	3.73
10	19-20	3.64	3.75	3.73
11	21-22	3.64	3.75	3.73
12	23-24	3.64	3.75	3.72
13	25-26	3.64	3.75	3.72
14	27-28	3.64	3.75	3.72

Figure 19: Third voltage sweep, abnormal group 1 negative voltage causing hard fault.

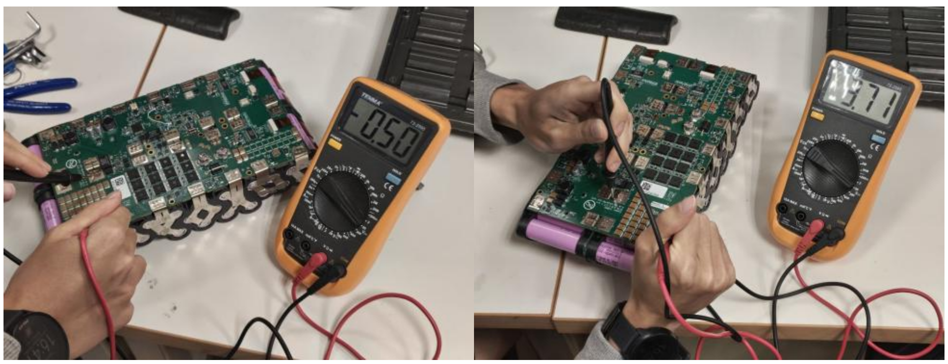

Figure 20: Group 1 reads a phantom −0.5V probing the pads but a healthy 3.71V across the cell terminals, confirming a broken negative-terminal weld.

4.3 Weld rework and verification

The pack had all its BMS welds undone and the packs separated for a detailed check to eliminate any further intermittent connections. After re-welding, the SoC fault lights cleared, the pack charged normally, and it reached a full, balanced charge for the first time (Figure 21). Battery module #1 was thereby repaired and verified.

Battery Information

Firmware Version	V47
State-of-Charge	98%
State	NORMAL
Voltage	55.410
Current	4.544
Log Capacity Used	2%
Recharge Cycle Count	95
CAN Node Serial Number	a2-21807-000c

Battery State

Cell Voltage Differential	0.06

Figure 21: First full charge with healthy battery balance index.

5. V2 pack design and validation

The major issue with design V1 is the unreliable spot welds from stacking nickel strips. Design V2 eliminates this issue and implements additional upgrades for a more robust construction.

A set of custom 0.15mm thick nickel tabs were designed to integrate seamlessly into the spacers, eliminating the nickel strip trimming and stacked-welding process in V1, which was time consuming and produced unreliable welds.

V2 added another set of braces with tighter tolerances (white) to hold the cell lattice rigidly, reincorporated the OEM injection-molded spacers for rigidity, and is re-dimensioned to fit the casing and PCB more accurately (Figure 22). They were printed in glass-filled ABS (ABS-GF) for dimensional stability at temperature.

One important design feature of the custom nickel tabs is the 0.8mm slot cutout at every weld site. The spot welder probes are positioned across the slot when welding, which encourages the spot welder's current to flow through the nickel layers instead of just shorting across the surface, producing much stronger welds with minimal sparking (Figure 23).

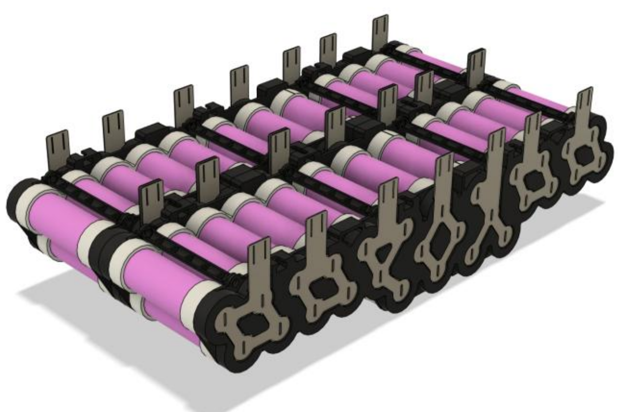

Figure 22: Design V2 adding braces (white), OEM spacers and custom nickel tabs.

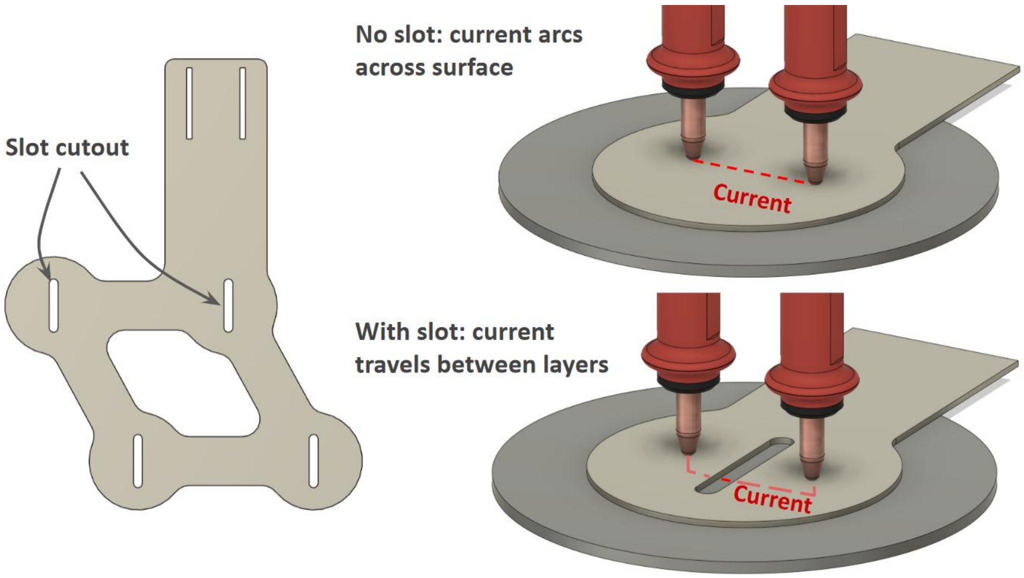

Figure 23: Illustration of slot cutout encouraging inter-layer current flow rather than shorting across the surface.

The improvement was borne out in assembly. After assembly, the SoC button was initially unresponsive; this was traced to a sheared SoC flat-flex cable (Figure 24). The weld quality achieved is visible in the completed pack (Figure 25). The broken cable was repaired by scraping back the insulation and soldering the exposed traces, with Kapton tape added as a strain relief on the joints. The SoC then read correctly. Pack #2 charged to full within about an hour, reported a very healthy balance index of **0.02** (Figure 26), and accepted a battery firmware update once installed in Spot. Both modules were thereby restored to a full, balanced charge, meeting the power-system objective.

Where pack #1 took two people three days and had multiple points of failure after assembly, pack #2 was completed by one person in a single afternoon and verified to have no faults, testifying the effectiveness of the design upgrades.

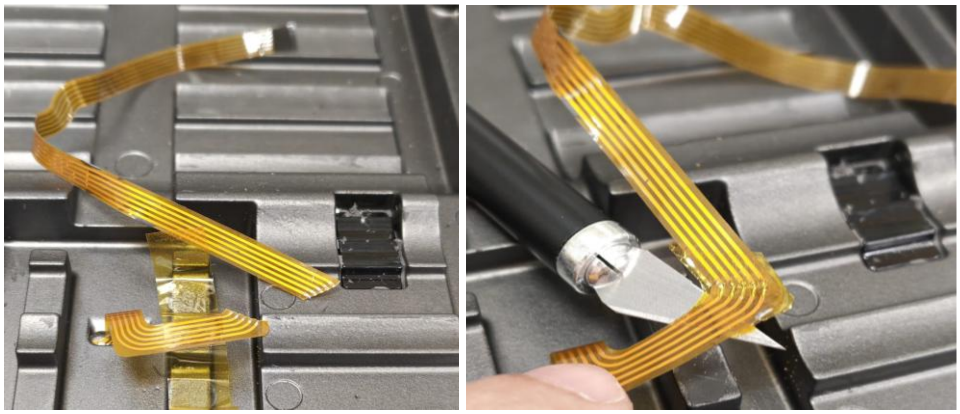

Figure 24: The sheared SoC flat-flex cable and the soldered, Kapton-secured repair.

Figure 25: Pack #2's laser-cut nickel strips perfectly welded in one pass.

Battery Information

Firmware Version	V47
State-of-Charge	97%
State	NORMAL
Voltage	55.827
Current	4.351
Log Capacity Used	2%
Recharge Cycle Count	41
CAN Node Serial Number	a1-20193-000c

Battery State

Cell Voltage Differential	0.02

Figure 26: Very healthy battery charge and balance index for pack #2.

6. Robot enablement and safety interlocks

6.1 DB25 payload-port cover detection bypass

Spot was received with one of the DB25 payload port covers missing. Enabling motor power raised an "uncovered payload port" stop function. Inspection showed the cover contained no active components, only a mating female DB25 in a shell, so the detection must be passive. Probing the cover for continuity revealed that four sets of pins are shorted together inside it. Replicating those shorts on the open port, first with tinned jumper wires (Figure 27), later with a female DB25 header wired to match (Figure 28), cleared the lockout and allowed motor power to be enabled. The pin-short map is given in Appendix H.

It was later discovered that Boston Dynamics has official documentation on the pinout of the DB25 payload ports[9], which confirms that the bench-tested pins to short are indeed the integrated safety loops.

Figure 27: Jumping payload port connector with wires to replicate cover.

Figure 28: DIY mating payload port connector with safety detection pins shorted.

7. Actuator mechanical investigation

With power restored, the controller prompted for a self-right procedure after Spot was powered on. In the initial phase of the self-right, Spot continuously fails to reach the "sit" pose; the left leg repeats a jerky motion as if trying to reach a specific position. When Spot's motors are powered on, they should default to a lock-out state; the *hl.hx* joint, however, remains limp did not resist back-driving. Furthermore, no motion was observed on the real-time 3D viewer in the console when the *hl.hx* joint was manually jogged around; it remains frozen at an extreme angle (Figure 29). Hence, attention was turned to the *hl.hx* actuator, specifically towards the angle sensing hardware.

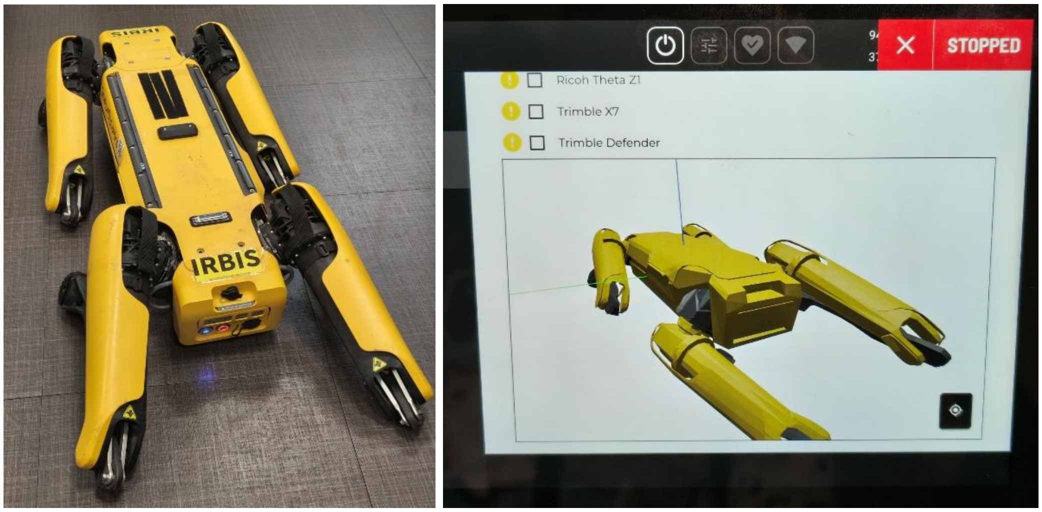

Figure 29: Spot's *hl.hx* joint is stuck frozen at an extreme angle while all other joints reflect accurately in the 3D viewer.

7.1 Teardown and component comparison

Spot was disassembled to extract the left hind leg (Figure 30). The full disassembly procedure is found in Appendix J. During teardown the leg wiring and controller board were inspected for physical defects and none were found. Hence, the investigation was framed around comparison with the healthy right hind leg.

Figure 30: Spot disassembled to isolate defective leg for testing.

Both hind legs were then removed and their *hx* motor driver boards compared side by side (Figure 31). The two drivers were physically identical, with no sign of component degradation on either. Rotating each motor by hand confirmed that the encoder's diametric rotor magnet was firmly seated, rotated freely, and stood at an identical protrusion height on both sides, ruling out an obviously displaced or loose magnet as the cause.

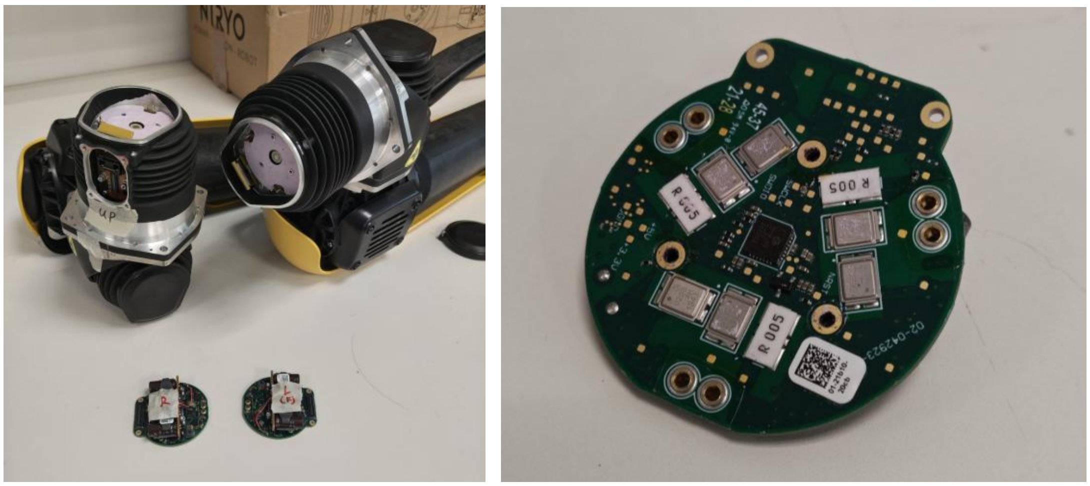

Figure 31: The two hind hx motor driver boards removed for direct comparison.

7.2 Driver and cable swaps

Two swaps tested the driver and the wiring. First, the left and right *hx* driver boards were exchanged (Figure 32) and the legs refitted; on power-up no motor faults were reported, but the abnormal joint behavior physically remained with the left socket.

Second, the *hx* motor cables were exchanged, after probing the connector PCBs to confirm they were not mirror images and that power and ground would map correctly. Again, the abnormal behavior stayed with the left socket. The 2 swaps exonerate motor driver and cable defects from the diagnosis.

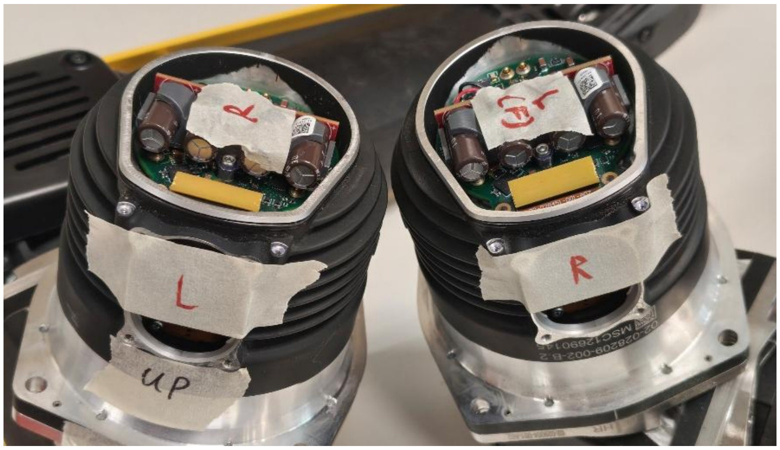

Figure 32: hx actuator drivers swapped.

7.3 Motor disassembly and actuator architecture

The *hx* actuators were further disassembled to identify any additional points of failure. The detailed disassembly process of the actuator is found in Appendix K. Upon disassembly, a secondary vertical radial iC-MU hall sensor (hereafter *output-stage encoder*) was discovered. The radial encoder reads off a Nonius magnetic ring attached to the actuator's harmonic drive output (Figure 33).

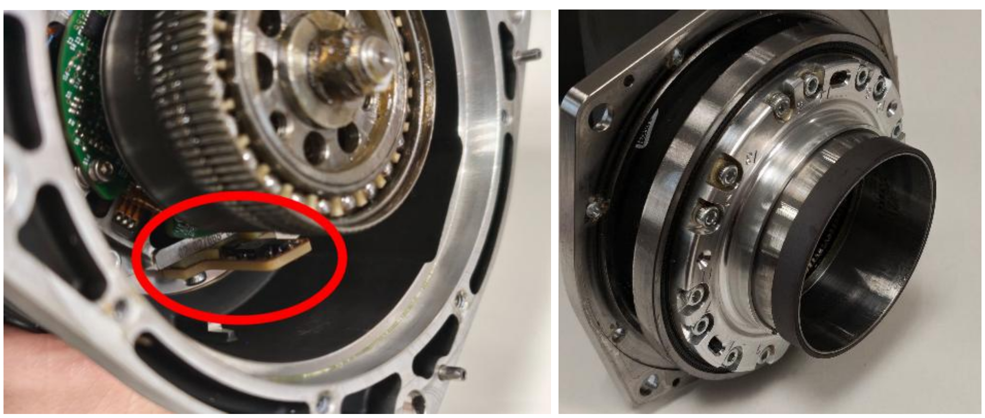

Figure 33: Secondary radial encoder (iC-MU) board and Nonius magnet ring on output.

The actuator's mechanical output is produced by a brushless motor turning through a 50:1, size 17 harmonic drive (Figure 34). The motor has 20 poles and 18 slots (Figure 35), with a phase resistance of approximately 0.5 Ω. The rotor assembly integrates a load cell array used for per-joint force feedback (Figure 36). The consolidated *hx* motor specifications are presented in Appendix L.

Figure 34: Size 17, 50:1 harmonic drive.

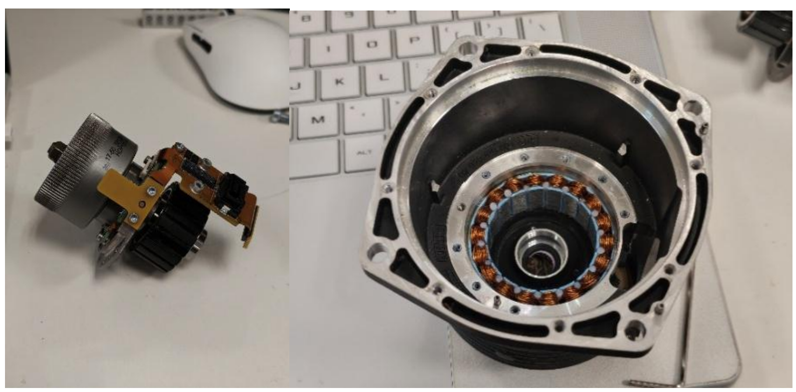

Figure 35: 20-pole rotor and 18-pole stator separated.

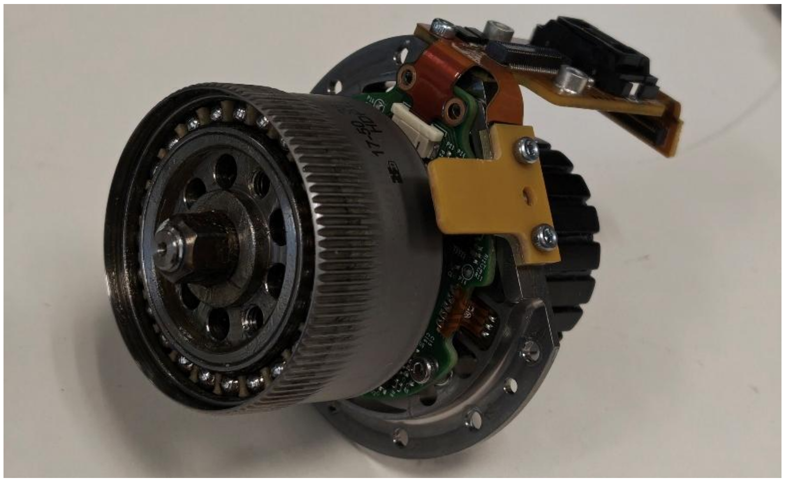

Figure 36: Integrated load cell array (green PCB under harmonic drive spline).

7.4 Attribution of fault to the output-stage encoder

Disassembly revealed a dual-encoder architecture not accounted for in the initial diagnosis. An axial encoder (iC-MHM) reads rotor angle at the input of the harmonic drive, and a second off-axis encoder (iC-MU) reads joint angle at the output.

The per-phase 5 mΩ shunt resistors (Figure 37) indicate in-line phase current measurement and therefore field-oriented control (FOC). The control algorithm implies a cascaded loop structure in which the rotor encoder serves the inner commutation and current loops via the Park transform[10], while the output encoder closes the outer joint-position loop.

Assigning the fault to the output encoder accounts for all previously documented symptoms under a single mechanism. The joint angle feedback to the admin console likely uses feedback from the output-stage encoder since the axial encoder only measured rotor orientation pre-reduction, which explains the frozen *hl.hx* in the admin console 3D view. When Spot's motors power on, the actuators latches its instantaneous joint position as the outer-loop setpoint to enter lock-out; a frozen feedback signal pins position error at zero and yields no corrective torque. This explains why the *hl.hx* joint is the only one back drivable at lock-out.

Diagnostic effort is therefore redirected to the off-axis encoder and its signal path.

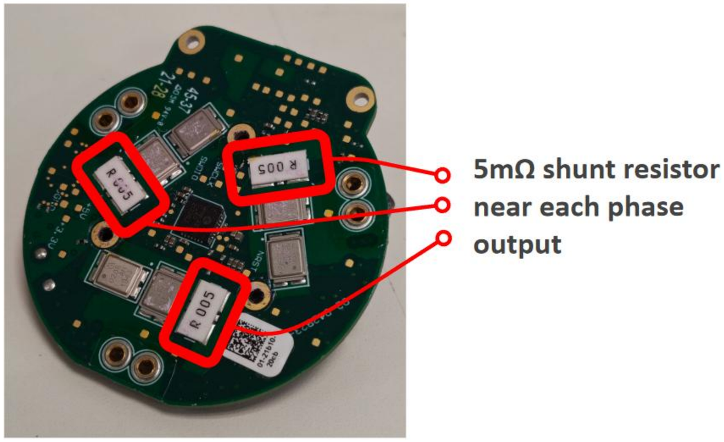

Figure 37: Per-phase current sense shunt resistors, strong indication of FOC implementation.

7.5 Output-stage encoder swap

The output-stage encoder module, which is one piece with the main connector PCB, was isolated and swapped between *hl.hx* and *hr.hx* (Figure 38). On power up, the frozen angle issue cleared and all joints were responsive as reflected in the 3D live view.

Figure 38: Output-stage encoder & connector PCB swap, highlighted in red.

8. Joint-level testing and calibration

8.1 Joint-angle readback and calibration

While the joints are responsive in the 3D live view, *hl.hx* and *hr.hx*'s reflected position is inaccurate, with *hl.hx* clipping into the upper angle limits (Figure 39). Each leg was jogged by hand across its full range of motion and the angles recorded at the physical extremes: "Min" and "Max" corresponds to the angle when the leg is fully tucked under or raised above Spot (Figure 40). The physical range (absolute angle delta) was consistent across all four legs, indicating that the offset was a fixed calibration shift.

The controller initially reported a "*hl.hx* joint angle out of range" hard fault that prevented the motors from even powering on. However, this could be bypassed by simply tucking the leg beneath the body before powering the motors so the *hl.hx* angle is within the acceptable range.

The joint calibration procedure was initiated after a successful power up (Figure 41). The initial phase of the calibration raises all joints to their physical limits and re-calculates an offset value, eliminating any further offset issues immediately.

Spot was able to move around with no issues after calibration concluded.

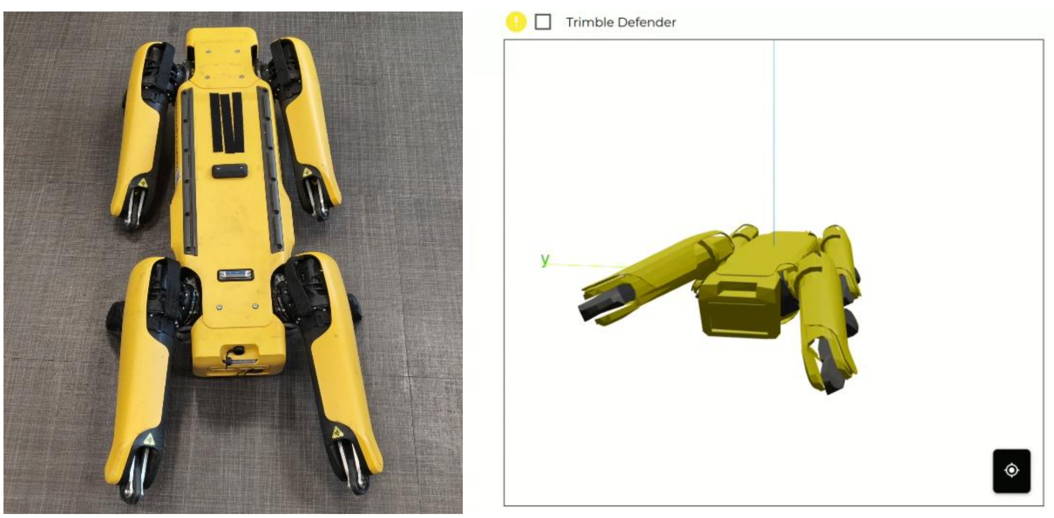

Figure 39: Spot's actual position (left) vs reported position in console (right). The *hl.hx* joint clips the upper angle limits in the viewer.

).

Hip angle (radians)		
	FL	**FR**
Min.	-0.789	0.789
Max.	0.787	-0.778
Delta	1.576	1.567
Center	-0.001	0.0055
	RL	**RR**
Min.	-0.252	0.905
Max.	1.303	-0.668
Delta	1.555	1.573
Center	0.5255	0.1185

Figure 40: Hip (hx) joint angles of all 4 legs recorded at the physical limits.

Figure 41: Spot performing joint calibration.

8.2 Frozen angle error returns after power cycle

However, when Spot was power cycled, the frozen angle issue returned, this time on the right hind hip (*hr.hx*). At the hardware level the swap sequence was conclusive in one specific sense: interchanging the secondary output-stage encoder relocated the frozen angle behavior from *hl.hx* to *hr.hx*, whereas interchanging the drivers and cables did not. The fault therefore resides in a physical hardware difference at the output stage encoder.

8.3 Output-stage encoder replacement and controlled reassembly

We suspect the output stage encoder from the *hl.hx* actuator is degraded after long periods of operation in the high temperature environment of the actuator. The encoder was desoldered from the flexible PCB and a new unit from Taobao was reflow soldered on using a hotplate (Figure 42).

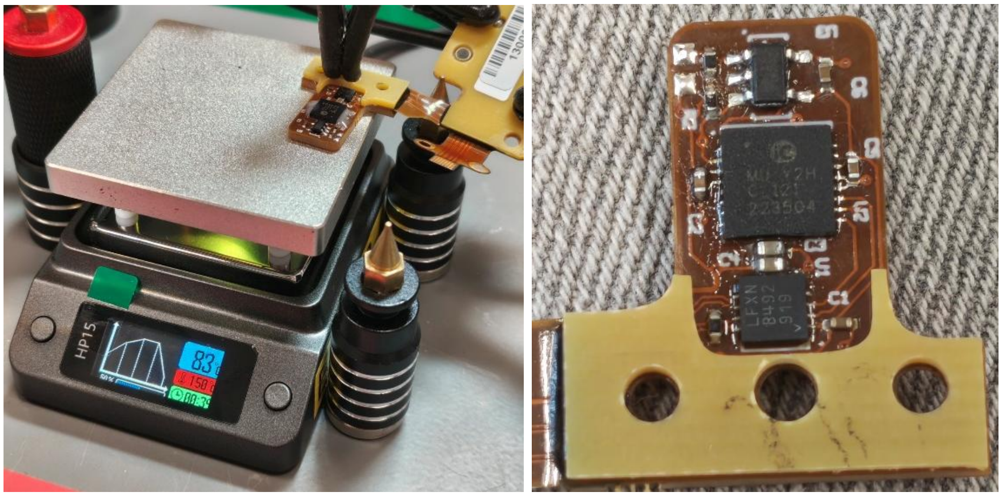

Figure 42: Reflow soldering the new iC-MU encoder (left), and close up of result (right).

We also hypothesized that mechanical interface between the angle-sensing components are rather sensitive, as the failure after a single successful operation may be attributed to the vibrations exacerbating misalignments within the actuator assembly. The reassembly process is thoroughly analyzed to identify any actions that may cause misalignment between the secondary encoder and Nonius magnetic ring.

The Nonius magnet ring used is dual track with 31 and 32 pole pairs on the Nonius and master track respectively [15]. With a recommended airgap of only 0.4mm and ring thickness of 6mm [15], alignment between the output stage encoder and Nonius ring is critical with small margins error margins.

2 potential points of misalignment were identified. Firstly, the output stage encoder board is retained by two screws with a single locating stub between them (Figure 43). As all holes are clearance-fit, the mount constrains translation only approximately and leaves the board free to rotate about the stub, so a horizontal installation is not guaranteed. Secondly, the eccentricity of the harmonic drive output with the Nonius ring attached is constrained by a large diameter cross-roller bearing clamped down by a flange plate (Figure 44); the screws on the flange has to be tightened equally to ensure vertical alignment of the Nonius ring and maintain the precise 0.4mm airgap.

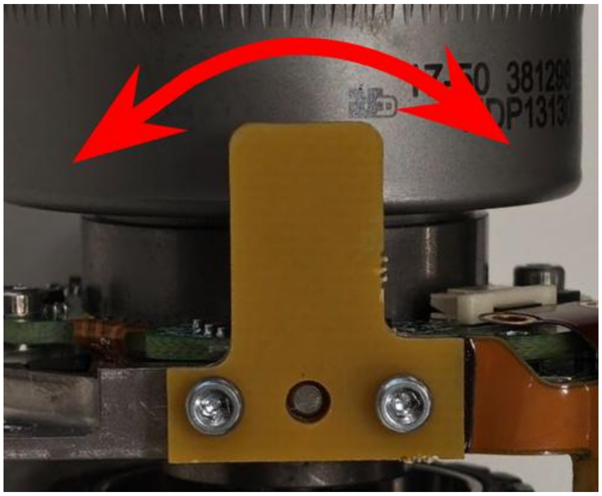

Figure 43: Clearance-fit mounting holes on the encoder PCB allows for angular play, affecting mounting precision.

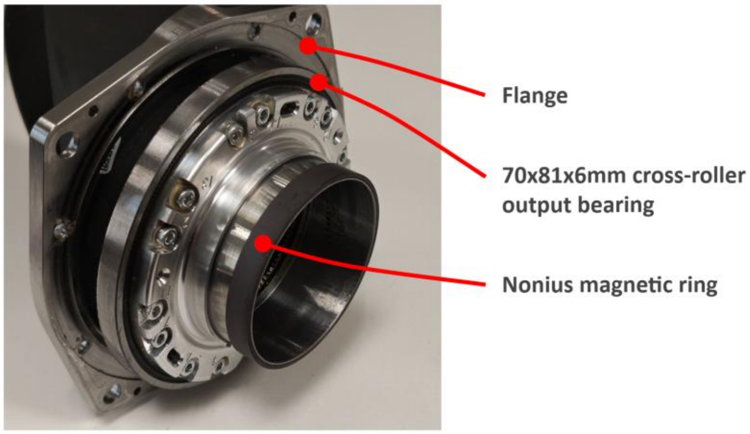

Figure 44: Flange used to constrain a rather thin output bearing.

The output stage encoder PCBs are swapped back to their original sides and the actuators reassembled with extra care taken to ensure consistency across the 2 units. The encoder PCB misalignment pitfall was circumvented by pressing both the PCB and rotor plate (which the PCB mounts to) on a flat surface before tightening the screws to ensure horizontal alignment. The actuator output assembly is inserted and pressed into the housing with evenly applied forces around the bearing. The flange screws are tightened using an electronic screwdriver with torque limit in a crisscross sequence to ensure even clamping force.

The frozen angle issues cleared once again after power up, and Spot was able to successfully perform another joint calibration.

8.4 Encoder health warning

While the calibration was successful, the calibration results reported an "Encoder Health <20%" warning for both the *hl.hx* and *hr.hx* actuators (Figure 45). Boston Dynamics does not have any official documentation on what this warning means specifically, but the console suggests returning the unit for repair.

This warning, however, does not impact robot operation. While teleoperating Spot to walk around, the console also intermittently shows warnings of "sensor misread" for both the *hl.hx* and *hr.hx* joints (Figure 46). However, the misread warning clears almost instantly and has no observable effects on Spot's motion; it was tested to still perform emotes and traverse staircases without issue with the warning present.

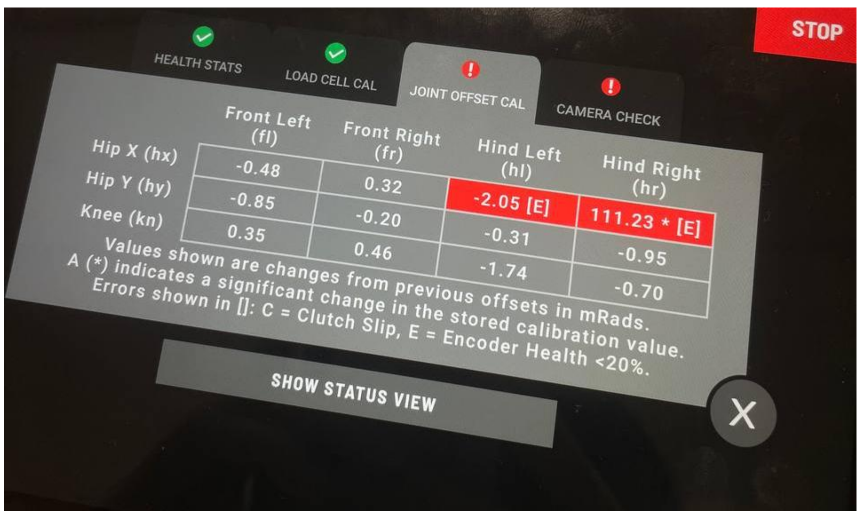

Figure 45: Calibration reports "[E] = Encoder Health <20%" warning for *hl.hx* and *hr.hx*.

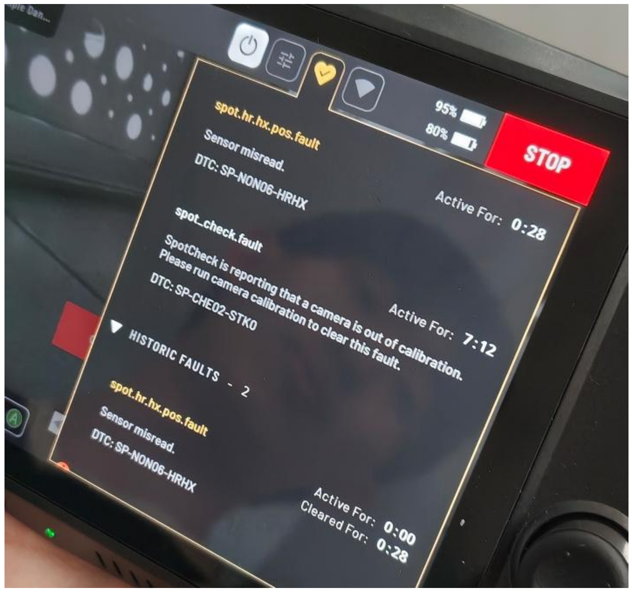

Figure 46: Intermittent “Sensor misread” warnings with no observable effects on operation.

8.5 Manual adjustment of encoder airgap and fault confirmation

Since the “encoder health” and intermittent “sensor misread” warnings are present even for the *hl.hx* joint with the freshly replaced iC-MU encoder, and mechanical alignment was done to the best of our abilities, attention was turned to the other angle-sensing component: the ferrite Nonius magnetic ring. We hypothesize the Nonius ring is degraded after years of operation in the warm actuator enclosure, causing it to lose magnetism and hold a weakly-defined field that the encoder struggles to track accurately.

The encoder is soldered on a piece of thin, flexible PCB that is sandwiched between 2 pieces of FR4 boards to provide rigidity for mounting. The FR4 boards were inspected and have no circuitry within. Hence, an attempt was made to manually adjust the encoder in *hl.hx* to sit closer to the Nonius magnet ring by filing the FR4 spacer material down by roughly 0.13mm (Figure 47). This is a substantial decrease from the original 0.4mm airgap [15].

After the encoder is reinstated and the legs reassembled, Spot was powered on and calibrated with spot check. The test result shows that the filed encoder no longer reports the “encoder health <20% error”, and the joint operated normally with no sensor misread warnings. This confirms the hypothesis that the frozen angle issues and occasional misreads were likely due to a degraded Nonius magnet ring which the encoder has trouble reading accurately.

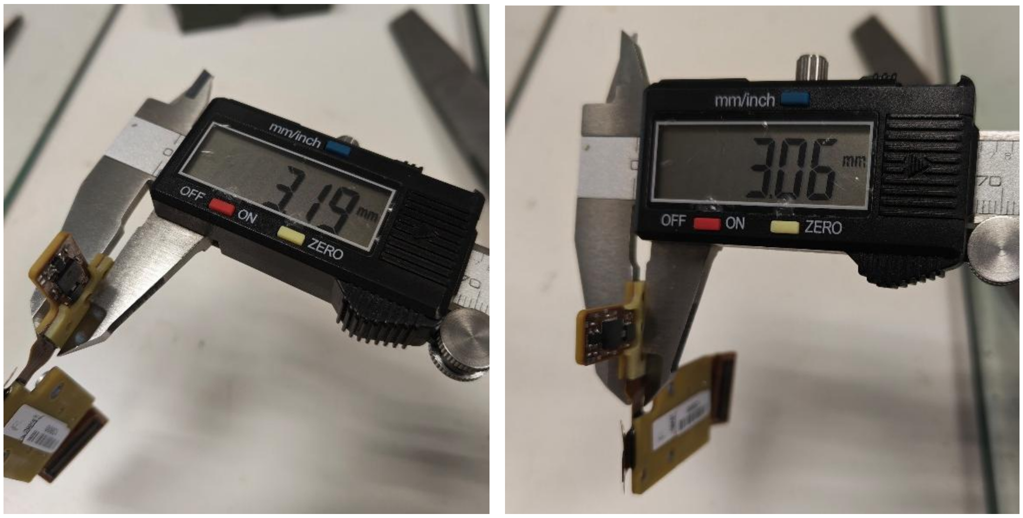

Figure 47: Encoder FR4 spacer width before (left) and after (right) filing.

9. Encoder electrical reverse-engineering

Component swaps, replacements and alignment adjustments localized the fault to a degraded ferrite Nonius magnetic ring, but the diagnosis is based on inference rather than measurement. Neither Boston Dynamics nor third-party sources document the full meaning of the error and warning messages or the electrical characteristics of the subsystems involved. The output stage encoder was therefore reverse-engineered to communicate with it directly and extract empirical data to verify the degraded ring hypothesis.

9.1 Board components and pinout

The secondary output-encoder PCB carries three principal devices: the **iC-MU** encoder itself, an external EEPROM holding its configuration, and an RS-485/RS-422 differential transceiver (Analog Devices LTC2863[12]) that carries the sensor's serial output off-board (Figure 48). A set of test pads exposes the EEPROM's SDA/SCL lines, presumably for programming, together with the transceiver's differential outputs and voltage supply near the main connector (Figure 49).

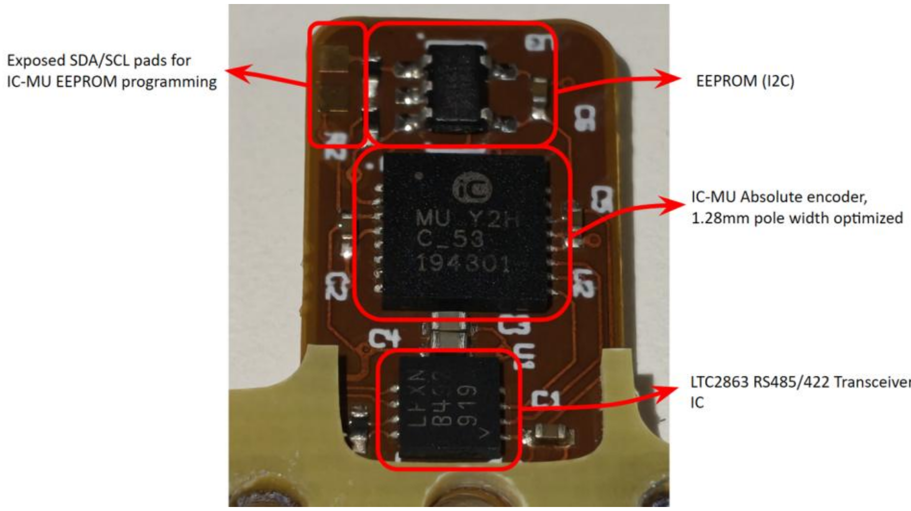

Figure 48: Component identification on iC-MU encoder module.

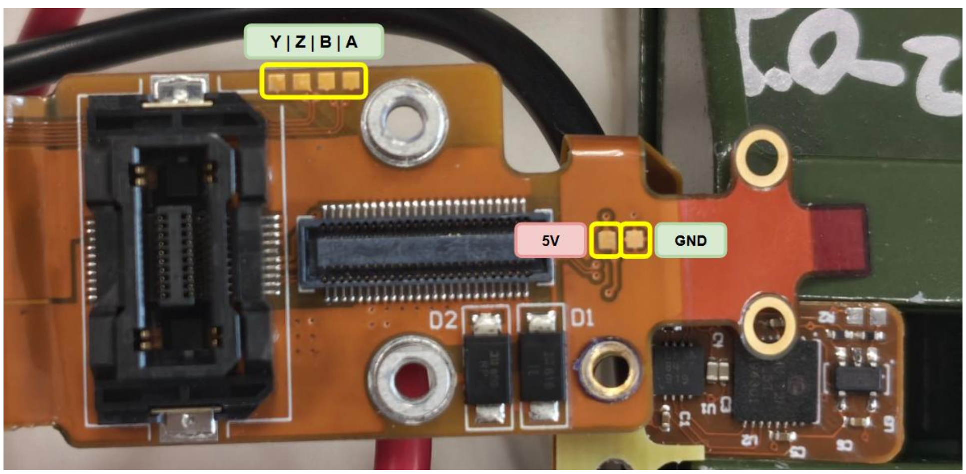

Figure 49: Test pads for transceiver lines, as well as 5V power located downstream close to the main connector.

Probing established the board's wiring as a textbook external SSI (ExtSSI) slave node according to the iC-MU datasheet[11]. The exact reverse-engineered connections and labelled pinouts are detailed in Appendix E, and the encoder-board pinout in Appendix G.

9.2 Configuration EEPROM dump

To characterize the encoder's configuration, we soldered wires directly to the EEPROM IC to supply power and connect to it via I2C. The EEPROM was provided power from the 5V pins on a Raspberry Pi 5, and the I2C pins were connected to the Raspberry Pi's hardware I2C pins (pin 3 and 5) (Figure 50). An I2C scan was performed and the EEPROM showed up with address 0x50. The i2cdump tool was used to dump all the bytes before formatting it as hex using xxd (hexdump). (Appendix F). The same process was repeated for the other encoder's EEPROM.

The contents of both EEPROMs are exactly identical when the extracted binary is diffed. The contents were then decoded against the iC-MU datasheet[11] to map the register values to the specific configuration.

The decode established that the stored configuration is a coherent, datasheet-normal ExtSSI set-up: amplitude control is enabled (manual gain fields ignored), Port A is configured as ExtSSI and Port B as ABZ, and the config-range CRC16 (0x52A8) matches the value computed over the configuration bytes. The full config map is found in Appendix F.

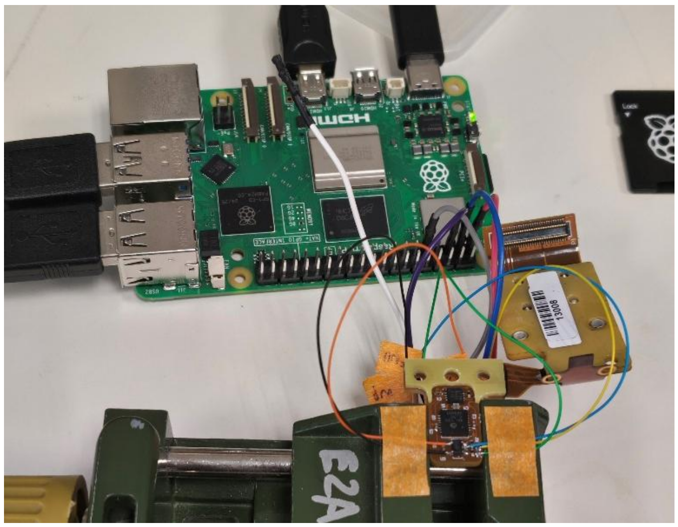

Figure 50: Dumping EEPROM over I2C using a Raspberry Pi.

The significance of this decode is that it eliminates configuration corruption as a possible fault: the board's programmed set-up is consistent between both the *hl.hx* and *hr.hx* boards and consistent with the datasheet-documented functions.

9.3 Signal routing

Probing to trace the sensor signals out to the main motor connector showed that no such route exists: none of the encoder transceiver's differential A/B/Z/Y outputs reach the main connector. Instead, the secondary output encoder is wired directly to the motor driver's STM32 microcontroller. We theorize that the driver STM32 consumes the post-reduction joint angle locally, formats feedback, and streams it to the main compute module over the driver's own CAN transceiver.

10. Encoder bench read attempt

The only empirical test that can pinpoint the issue is a direct read to monitor the sensor's raw output from the output stage encoder while it is seated in the enclosed actuator and tracking the Nonius magnetic ring. Extension wires were soldered to the differential signal pairs and power connection test pads mapped in Figure 49 for accessibility (Figure 51).

2 options are presented for communicating with the sensor: using a full duplex RS422 transceiver / 2x serial-to-RS485 transceivers, or attempting to use the driver STM32 to generate the data acquisition pulses. The former is infeasible as the required hardware was not readily available and there was insufficient time to purchase it online. The feasibility of the latter was also obscure as there was no documentation on the STM32 behavior. The assumption for the 2nd test is that the STM32 is programmed to continuously request sensor data, hence eliminating the need for us to initiate requests manually. Instead, the output (Y/Z) differential lines can be hooked up to an oscilloscope/logic analyzer to capture the data returned by the encoder.

Test pads labelled +5V and +3.3V were identified on the driver PCB (Figure 52), and probing reveals the +5V is continuous with the voltage supply to the encoder. We theorize that the +5V rail feeds into the on-board 3.3V LT3029 LDO[13] which powers the STM32. However, when we hooked up the +5V to a power supply, the +3.3V test pad only measured 0.4V likely due to the enable (EN) pin on the LDO being pulled low; instead, we hooked the +3.3V test pad directly to another power supply and the STM32 was successfully powered, With power connected and the transceiver's A/B lines to an oscilloscope, no edges were detected, meaning the STM32 was not actively sending requests to the sensor (Figure 53). There were also no edges detected when the oscilloscope was connected to the Y/Z output lines, marking the test as a failure.

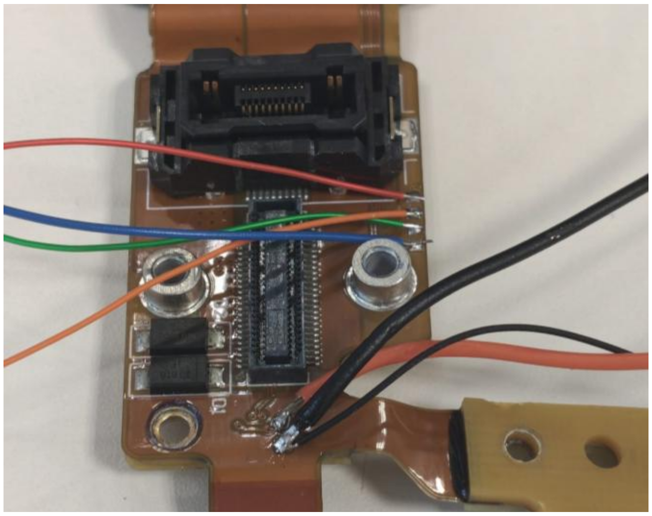

Figure 51: Extension wires soldered to power and transceiver differential pairs.

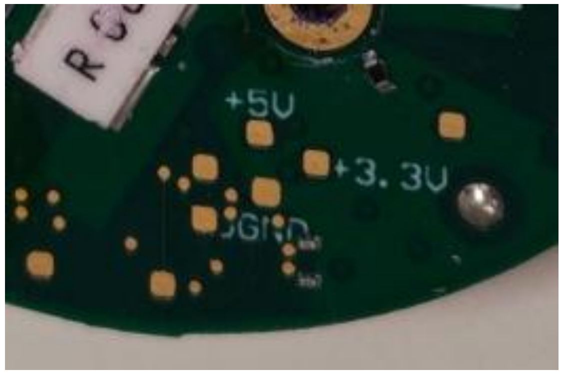

Figure 52: +5V and +3.3V test-pads on the motor driver. Only 0.4V was detected from the +3.3V test pad when 5V was supplied.

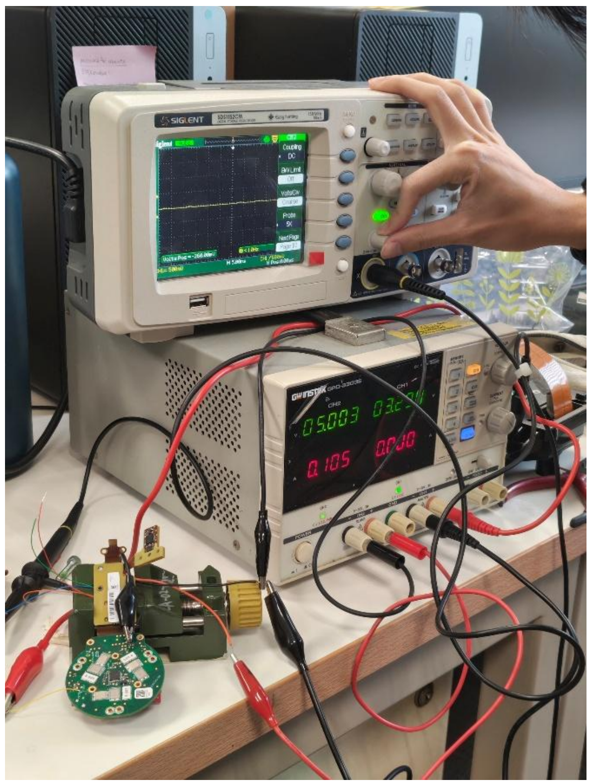

Figure 53: No edges detected on transceiver differential lines with STM32 powered.

11. Evaluation, limitations and improvements

11.1 Power system

The power-system objective was met in full: both modules were rebuilt on their original BMS boards in the OEM 14s4p form factor and both now reach a full, balanced charge, pack #2 at a balance index of 0.02. The governing finding was that reliability depended almost entirely on spot-weld quality — the same class of defect appeared first as cell-group imbalance and later as a battery-fault lockout, both traceable to bends and stress points in the nickel-strip routing imposed by the V1 bracket. V2 turned that into a design change: added bracing, OEM-derived spacer positions, and laser-cut slotted strips that avoid stacking. It welded cleanly, balanced on the first attempt, and assembled in a fraction of the time. The only limitation remaining is the lack of a phase change material (PCM) filler akin to the OEM pack for thermal buffering, leaving the pack reliant on the retained NTC monitoring.

11.2 Actuator, at the hardware level

Comparative swapping between the faulty *hl.hx* and healthy *hr.hx* actuators ruled out the driver electronics, the motor wiring, the rotor and its magnet, and the primary rotor encoder, since none of those parts carried the abnormal behavior with it; only interchanging the secondary output-encoder PCB moved the fault between joints (Appendix D). Electrical reverse-engineering narrowed that difference further: the board is a datasheet-normal ExtSSI iC-MU node whose stored configuration decodes as CRC-valid and identical on both joints, eliminating configuration corruption and pointing to a device-level difference. The fault was finally attributed to the ferrite Nonius ring the encoder reads: filing roughly 0.13 mm off the FR4 spacer to close the airgap cleared the "encoder health <20%" error and the intermittent sensor misreads, and the joint has operated normally since, consistent with a ring that lost field strength in service. That attribution rests on the joint's response to the airgap correction rather than on a direct measurement of the ring's remanence, which was not attempted.

The main unfinished work is the in-situ bench read of the iC-MU's raw registers, blocked by two independent obstacles: no suitable full-duplex RS-485/RS-422 interface was available, and the driver STM32 does not generate the request signals periodically. Procuring the transceiver would close the issue.

12. Conclusion

The hardware objectives of the project were met. Spot's power system was rebuilt from two dead modules into two modules that charge to a full, balanced state, by replacing the degraded cells with OEM-spec Samsung INR18650-30Q cells on the original BMS and, critically, by driving the rebuild's reliability through spot-weld quality and a second-generation bracket design. In parallel, the faulty *hl.hx* actuator, a previously undocumented piece of hardware, was characterized from the outside in: its 50:1 harmonic-drive, dual-encoder construction was established by teardown; a disciplined sequence of component swaps localized the fault to a physical difference at the secondary output-encoder board; and reverse-engineering that board established its ExtSSI iC-MU design, pinout, direct-to-STM32 routing and CRC-valid configuration, narrowing the fault to a physical device difference rather than a programming or wiring one.

The skills developed span the rebuild and characterization of lithium battery systems and their BMS, the mechanical teardown and reassembly of a precision harmonic-drive actuator, and the electrical reverse-engineering of a Hall-encoder PCB, including datasheet-level EEPROM decoding and pinout tracing. The principal thread of future work follows directly from the one hardware task that could not be completed: instrumenting the iC-MU sensor with the correct RS-

485/RS-422 interface to read its raw output on the bench. Beyond that, a physics-based simulation of the actuator and its encoder signal chain is proposed as a means of exploring the fault without repeated teardown; it is noted here strictly as **planned** work, no such simulation having been built within this project.

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13. Appendices

Appendix A: Project hours log

13 May 2026	Wed	Battery controller inspection & charge test	Battery	3
20 May 2026	Wed	Battery restoration attempt (cell trickle-charge)	Battery	5
21 May 2026	Thu	Battery module complete disassembly & cell sourcing	Battery	7
25 May 2026	Mon	Battery pack design research (Li-ion safety)	Battery	4
25 May 2026	Mon	CAD design of battery spacer	Battery	8
28 May 2026	Thu	Replacement battery assembly (1) — spot-weld & BMS test	Battery	8
29 May 2026	Fri	Replacement battery assembly (2)	Battery	6
02 Jun 2026	Tue	Replacement battery assembly (3)	Battery	8
04 Jun 2026	Thu	Replacement battery assembly (4)	Battery	8
05 Jun 2026	Fri	Battery firmware update & Spot power-up test	Battery	5
11 Jun 2026	Thu	Replacement battery assembly (5) — weld fix & recharge	Battery	6
12 Jun 2026	Fri	Joint diagnosis (actuator fault, data collection)	Motor	6
15 Jun 2026	Mon	Battery pack #2 & Spot teardown	Motor	6
16 Jun 2026	Tue	Battery pack #2 design optimization	Battery	4
17 Jun 2026	Wed	Battery pack #2 assembly at Sodion	Battery	5
18 Jun 2026	Thu	Battery #2 SoC cable repair & driver diagnostics	Motor	7
19 Jun 2026	Fri	ABA leg component diagnostics (component swap)	Motor	6
25 Jun 2026	Thu	Leg motor assembly teardown	Motor	4
30 Jun 2026	Tue	Output encoder PCB swap	Motor	6

01 Jul 2026	Wed	SpotCheck diagnosis + magnet swap	Motor	8
02 Jul 2026	Thu	Encoder EEPROM probe (iC-MU I2C bench rig)	Motor	8
14 Jul 2026	Tue	Encoder chip swap & FX23 breakout bring-up	Motor	6
15 Jul 2026	Wed	Encoder board swap L↔R (third SpotCheck)	Motor	6
16 Jul 2026	Thu	Encoder register read attempt (RS485 tap)	Motor	6
17 Jul 2026	Fri	Left hip-X air-gap fix; rear camera repair	Motor	6
22 Jul 2026	Wed	Autonomy tests + actuator-repair validation	Motor / Autonomy	7
22-26 Jul 2026	Wed-Sun	Write report	Report	15
			TOTAL	**174**

Table 2: Project hours log

Appendix B: Justification for omission of phase-change filler material

The pack comprises 56 Samsung SDI INR18650-30Q cells operating at a peak cell current of 2.63 A and an RMS cell current of 1.50 A over a maximum mission duration of 1.5 hours (Appendix C). Ohmic heat generation has been evaluated using a DC internal resistance of 45 mΩ rather than the 26 mΩ figure quoted in the cell specification [16], as the latter is an AC 1 kHz upper-limit measurement that captures only the ohmic component of cell impedance and excludes charge-transfer and diffusion contributions present under DC load.

On this basis, peak and continuous dissipation are 0.31 W and 0.10 W per cell respectively, giving a pack-level continuous heat load of 5.7 W. Three independent arguments support the adequacy of passive dissipation without phase-change filler.

1) The cell specification qualifies the INR18650-30Q for 15 A continuous discharge at 25 °C with no forced cooling specified [16]; the present duty represents 1.0 % of the qualified continuous heat generation rate and 3.1 % at peak.

2) The 1.5-hour mission duration renders this a bounded transient rather than a steady-state problem: taking the cell mass of 48 g [16] and a specific heat capacity of approximately 1.0 J/g·K gives a cell thermal capacitance of 48 J/K, so under a fully adiabatic assumption, the cell self-heating over the complete mission is bounded at approximately 11 K.

Appendix C: Nickel strip sizing justification

The pack is rated at about 500 Wh and a datasheet runtime of roughly 1.5 h. Assuming discharge to 20 % state of charge over that time, the average power is approximately 0.8 × 500 / 1.5 ≈ **267 W**; against the manual's 38-52 V supply this implies an operating current of about **5.1-7.0 A**. Sizing to a 50 % margin (~400 W continuous) gives a peak pack current of about **10.5 A**, or roughly **2.63 A per cell** across the four parallel cells in each group. Those figures set the strip widths in the table below; the OEM pack uses 12 mm strip for each group output and 5mm strips between cells, which is comfortably conservative for the actual current.

Conductor	Spec	Basis
Inter-cell links	0.15 × 8 mm pure nickel	Rated 11.33A optimal [17], ample for per-cell peak current ~2.63A
4s pack leads	0.2 × 12 mm pure nickel	Rated ~22.67 A optimal / 34 A acceptable [17], ample for ~10.5 A pack max

Table 3: Nickel strip conductor sizing and basis

Appendix D: Consolidated swap tests and observations

Physical action	**Physical observation**
Exchange L/R hx driver boards	No motor faults on power-up; abnormal behavior stayed with the left socket, not the board.
Exchange L/R hx connector cables	Abnormal behavior stayed with the left socket; the other joint responsive and normal.
Exchange output-encoder PCB only (rotors/load-cells returned to original sockets)	Abnormal frozen behavior cleared on both joints; a residual mechanical offset remained on one joint. However frozen angle issue reappeared upon power cycle, relocated to the right socket.

Table 4: Consolidated swap tests and observations

Appendix E: Component pinout and connections on the iC-MU PCB

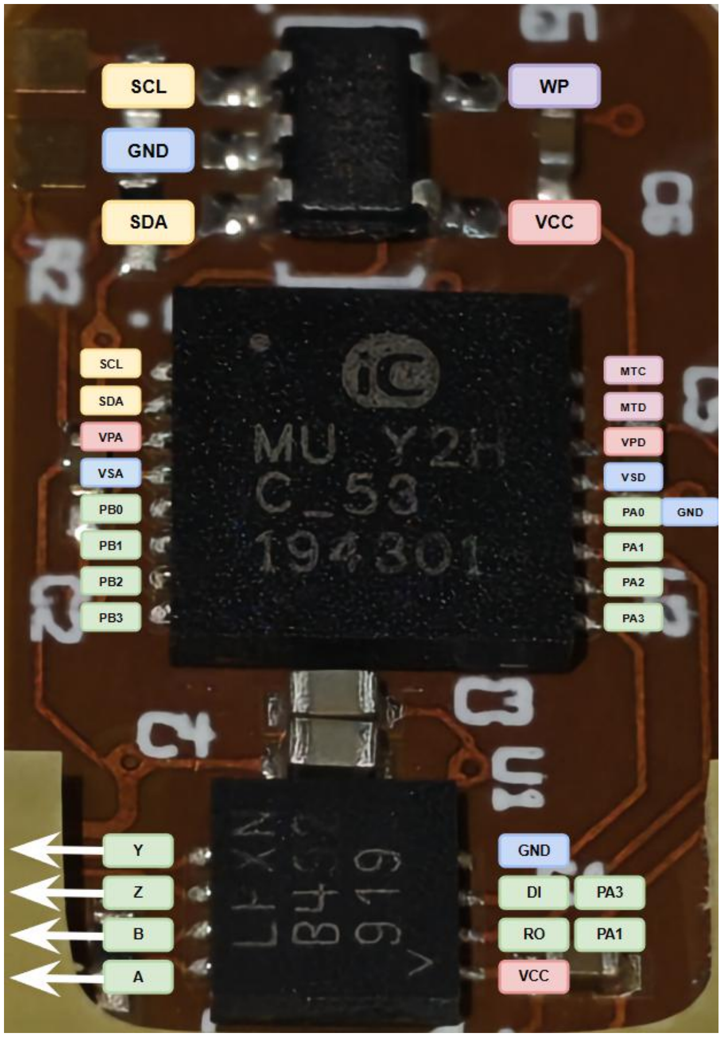


Figure 54: The iC-MU pinout and connections, wired as an ExtSSI slave node.

Appendix F: iC-MU EEPROM configuration register map

Full annotated decode of the external configuration EEPROM against the iC-MU datasheet [11]. Raw dump (hex), addresses 0x00-0x3F:

```
00000000 00 00 00 00 00 88 00 00 00 00 00 07 00 00 14 05
00000010 00 4e 20 ff 0f 13 10 02 00 00 00 00 00 00 00 00
00000020 00 52 a8 ff ff ff ff ff ff ff ff ff ff ff ff ff
00000030 00 00 00 00 00 00 00 00 4d 55 07 00 00 00 69 43
```

Addr offset	Raw	Register / fields	Decoded setting
0x00	00	GC_M, GF_M	Master coarse/fine gain 0 (ignored,

Addr offset	Raw	Register / fields	Decoded setting
			amplitude control enabled).
0x01	00	GX_M	Master cosine gain adjust = 1.000.
0x02	00	VOSS_M	Master sine offset = 0 mV.
0x03	00	VOSC_M	Master cosine offset = 0 mV.
0x04	00	PH_M	Master phase adjust = 0°.
0x05	88	ENAC=1, CIBM=8	Amplitude control active; bias-current trim nominal 0%.
0x06	00	GC_N, GF_N	Nonius coarse/fine gain 0 (ignored).
0x07	00	GX_N	Nonius cosine gain adjust = 1.000.
0x08	00	VOSS_N	Nonius sine offset = 0 mV.
0x09	00	VOSC_N	Nonius cosine offset = 0 mV.
0x0A	00	PH_N	Nonius phase adjust = 0°.
0x0B	07	MODEA=7, MODEB=0	Port A = ExtSSI (NPRES/MA/SLI/SLO); Port B = ABZ (A/B/Z/NER).
0x0C	00	CFGEW	All error/warning messages enabled.
0x0D	00	ACC_STAT, NCHK_CRC, NCHK_NON, ACRM_RES, EMTD	Actual (non-accumulated) status; cyclic CRC + nonius verification active; no auto-reset; min error-display time 0 ms.
0x0E	14	LIN=1, FILT=4	Linear/radial scanning; FILT4 = 39 dB suppression, 14-bit interpolation.
0x0F	05	MPC=5	32 master / 31 nonius periods; ≤19-bit

Addr offset	Raw	Register / fields	Decoded setting
			absolute resolution.
0x10	00	*_MT	No external multiturn data; MT chain/verification disabled.
0x11	4E	OUT_ZERO=2, OUT_MSB=14	Serial output inserts 2 zero bits; output MSB = user-data bit 27.
0x12	20	MODE_ST=2, …	Binary output, no SSI ring; raw master/nonius track data; LSB = bit 0.
0x13-0x14	FF 0F	RESABZ=0x0FFF	ABZ / FlexCount resolution = (0x0FFF + 1) × 4 = 16 384 edges.
0x15	13	SS_AB=1, FRQAB=3, …	AB step 1; startup counting visible; AB limit ≈ 781.25 kHz (320 ns edge distance).
0x16	10	LENZ, CHYS_AB=1, PP60UVW, INV_*	Z index = 90°; AB hysteresis 0.175°; UVW 120°; A/B/Z not inverted.
0x17	02	RPL=0, PPUVW=2	Config mode, no restriction; UVW commutation = 1 pole pair.
0x18	00	TEST	Normal mode.
0x19-0x20	00	SPO_BASE, SPO_0…14	Nonius track-offset slopes all zero.
0x21-0x22	52 A8	CRC16	0x52A8, matches CRC over config ranges 0x00-0x20, 0x30-0x3F.
0x23-0x27	FF	OFF_ABZ	Absolute-output offset block reads erased (unused).
0x28-0x29	FF FF	OFF_UVW	UVW offset reads erased.

Addr offset	Raw	Register / fields	Decoded setting
0x2A-0x2E	FF	PRES_POS	Preset position reads erased.
0x2F	FF	CRC8	Does not match posted offset/preset bytes (block unused).
0x30	00	PA0_CONF	Falling edge on NPRES = NO_FUNCTION.
0x31	00	EDSBANK	No EDS.
0x32-0x33	00 00	PROFILE_ID	0x0000.
0x34-0x37	00	SERIAL	0x00000000.
0x38-0x3D	4D 55 07 00 00 00	DEV_ID	ASCII “MU”, then 0x07 00 00 00.
0x3E-0x3F	69 43	MFG_ID	0x6943 = ASCII “iC” (iC-Haus).

Table 5: iC-MU EEPROM configuration register map

Appendix G: Encoder-board pinout

See Figure 54 and the pinout table in Appendix E for the ExtSSI wiring, and Figure 55 for the connector-flex test pads (the transceiver’s A/B/Y/Z differential pairs plus +5 V and ground) used for the bench read attempt of Section 10.

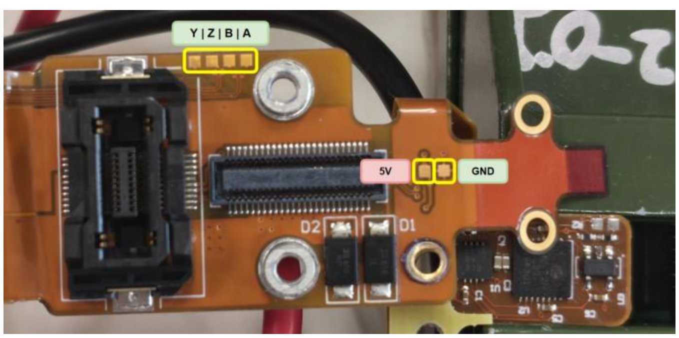
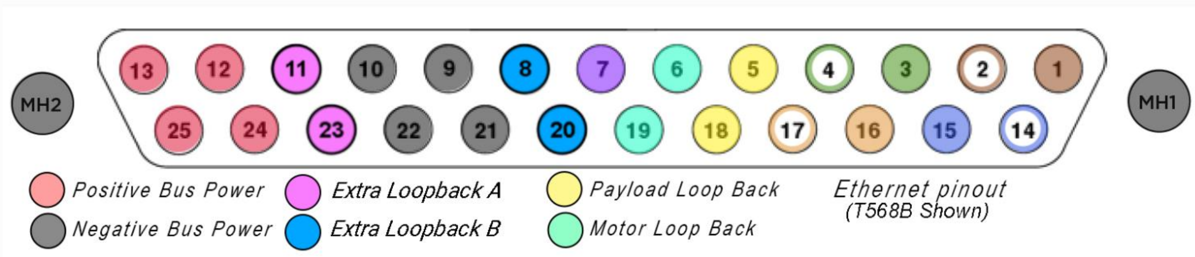


Figure 55: The connector-flex test pads, mapping to the transceiver’s A/B/Y/Z outputs and the encoder supply rails.

Appendix H: DB25 payload-port cover pin shorts

The four sets of pins shorted inside the OEM payload-port cover, reverse-engineered by continuity probing and replicated on the open port to bypass the cover-detection lockout (Section 6).

Figure 56: The DB25 payload-port cover pin shorts. Replicating these four shorts on the open port clears the “uncovered payload port” motor-power lockout.



Payload port pinouts by category

Category	Pins	Specifications
Power	12, 13, 24, 25	Voltage supply range: 35-59V Absolute Maximum Voltage: 72V Max current: 3A/pin Max power: 150W/port Bulk capacitance: 150uF/port
Communication	1-4, 7, 14-17	Ethernet: 1000Base-T PPS Accuracy: 5ppm PPS Frequency: 1Hz
Safety Loops	5 to 18, 6 to 19, 8 to 20, 11 to 23	Payload power interlock Motor power interlock Extra interlock loop (do not remove) Extra interlock loop (do not remove)

Figure 57: Payload port pinmap from Boston Dynamics, safety loops match bench-tested pins.

Appendix I: OEM cell spacing drawing

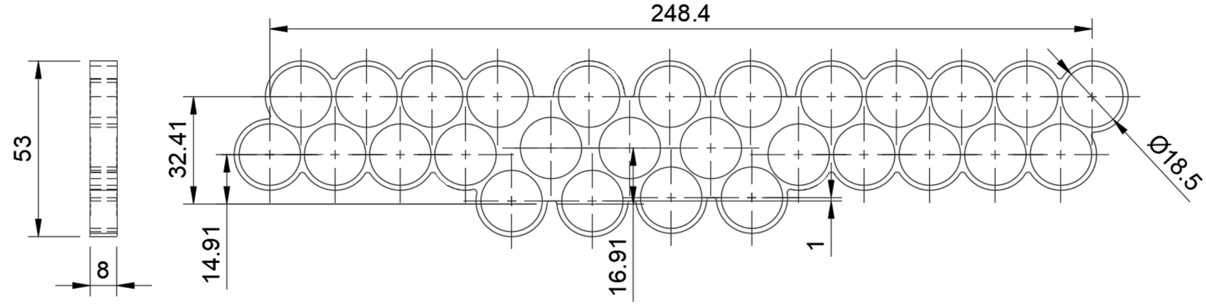
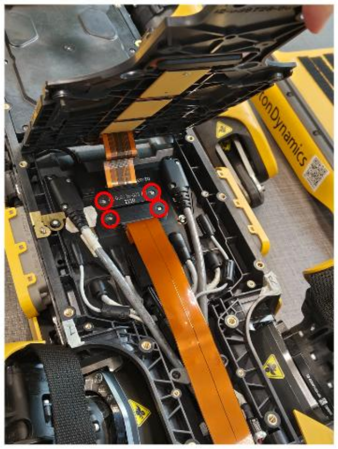


Figure 58: OEM cell spacing CAD drawing

Appendix J: Spot teardown and leg-extraction procedure

Remove 12x screws on rails (red), 6x screws from cover (blue) and payload port cover (green)

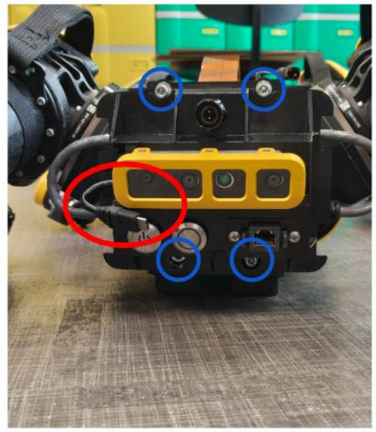
Remove top cover



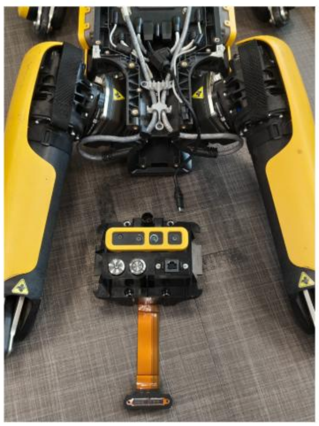
Remove 4x screws holding down payload interface/rear connectors

The rear yellow faceplate can be removed by undoing 4x metal clips on the underside, circled in red

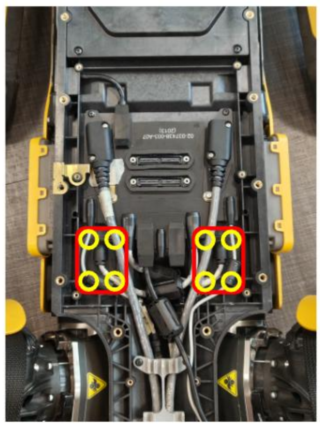
Remove the rubber seal



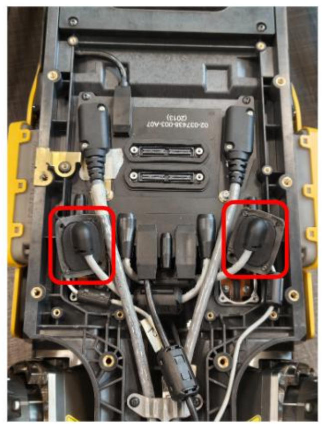
Detach rear faceplate, unplug rear camera USB-C cable (red) and remove 4x screws (blue)



Remove rear panel



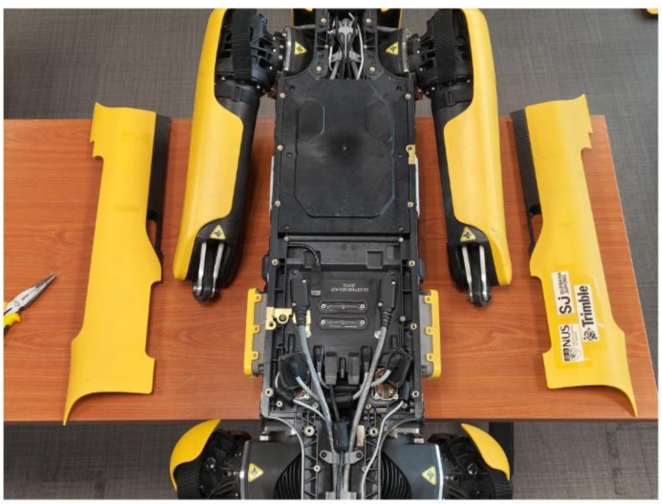
Remove both hx motor connectors (red), each held in by 4x screws (yellow)



Pull hx motor connectors vertically up to unplug

Remove both hy motor connectors (red), 4x screws for each (yellow)

hy motor connectors removed

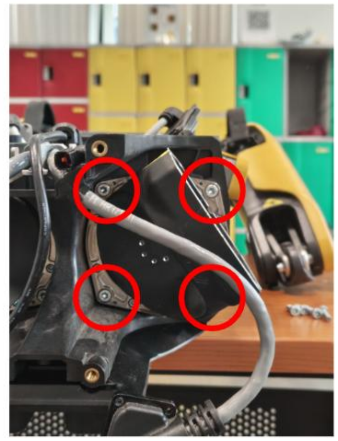


Remove side panels. Panels are attached with velcro, pull hard to release

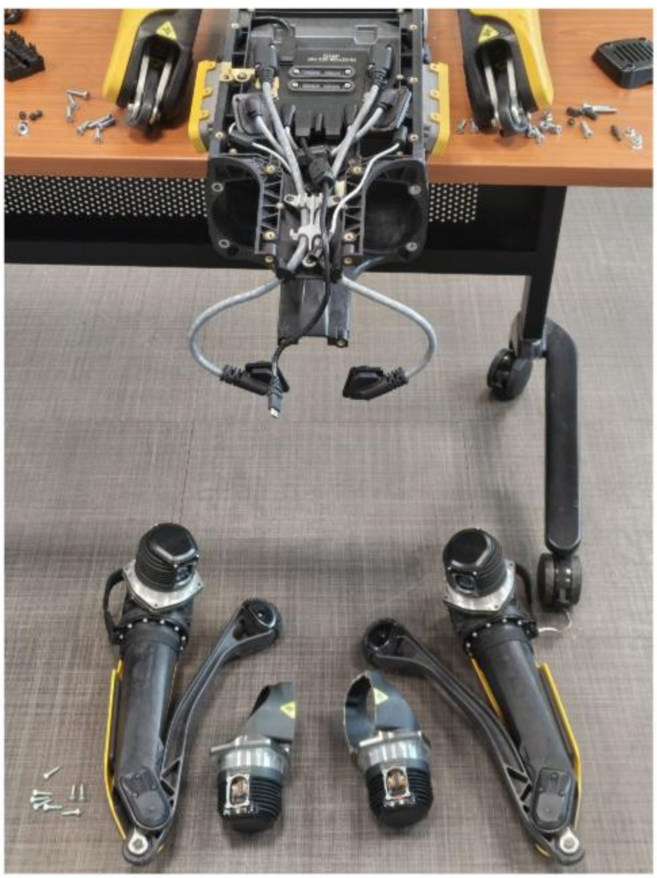
Remove both hy motor fan covers under the leg, 4x screws each

Remove 4x screws on the hy motor flange to detach legs below the hips.

Legs detached, leaving hx motor on body

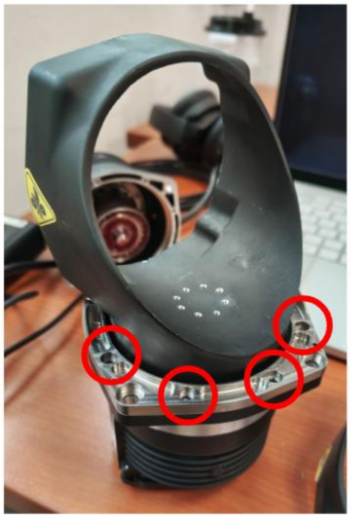


Remove both hx motors by unscrewing 4x screws on each hx motor flange. Note the screws are secured by flanged nuts on the opposite side



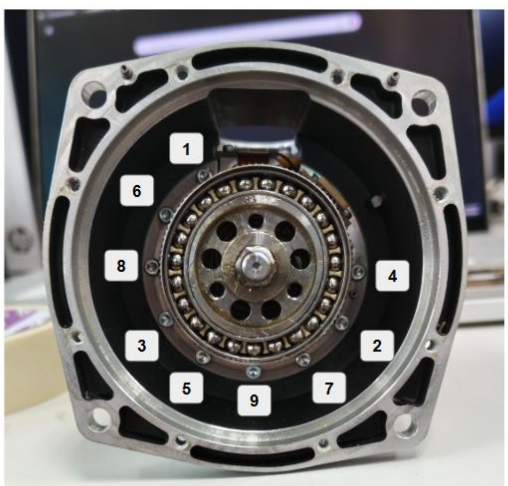
Both legs disassembled

Appendix K: hx actuator disassembly procedure

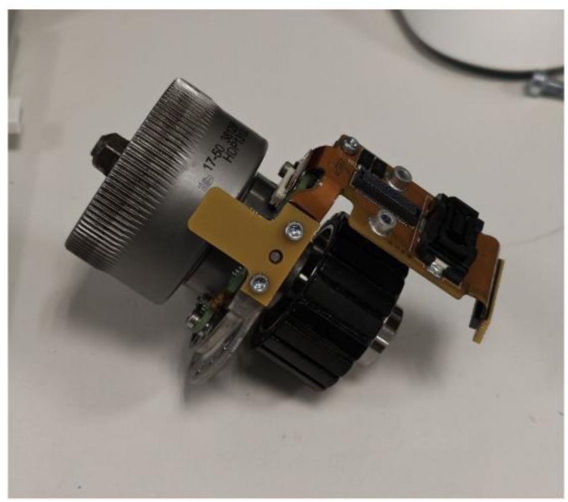


Remove 8x screws from the hx motor flange

The actuator can be pulled apart with some force; note that even, vertical pulling force should be applied to not damage the output encoder.



Unscrew 9x screws holding down rotor assemble. Note the screws are labelled and should be tightened in ascending order during reassembly



Rotor fully removed

Appendix L: Consolidated hx motor specifications

Motor and drivetrain parameters measured or observed during the actuator disassembly described in Section 7.

Parameter	Value	Basis
Motor type	Three-phase BLDC, field-oriented control	Per-phase 5 mΩ shunts and driver topology
Pole / slot count	20 poles, 18 slots	Direct count on stator and rotor
Phase resistance	≈ 0.5 Ω	Measured across motor phases
Gearing	50:1 harmonic drive, size 17	Identified on disassembly
Force feedback	Load-cell array integrated into rotor assembly	Per-joint, calibrated to socket
Output-stage sensing	iC-MU radial Hall encoder, Nonius magnetic ring	Nonius ring mounted to harmonic-drive output

Table 6: Consolidated hx motor specifications

Appendix M: Project expenditure

Components were procured against a single shared project budget across the two-member team, so the cost record is reported jointly. All amounts are in Singapore Dollars (SGD).

No.	Item / Part No.	Supplier	Unit price (S$)	Qty	Subtotal (S$)
Item Request List 1 — Power-system rebuild					
1	INR18650-30Q lithium cells	FalconPEV	4.50	112	504.00
2	DB25 male connector	Ask Electronics (Sim Lim Tower)	2.50	1	2.50
3	Custom laser-cut 0.2 mm nickel sheet	Taobao	83.32	1	83.32
Subtotal, power-system rebuild					**589.82**
Item Request List 2 — Actuator and sensor repair					
1	FX23-20P encoder breakout	Taobao	11.27	1	11.27

No.	Item / Part No.	Supplier	Unit price (S$)	Qty	Subtotal (S$)
	connector (Hirose)				
2	iC-MU secondary-encoder chip	Taobao	37.50	1	37.50
3	SK2 harmonic grease, 100 g	Taobao	20.99	1	20.99
4	Assorted Torx screws	Taobao	5.53	1	5.53
5	RTV silicone, black	MR FIX	6.90	2	13.80
6	FUJIDA multipurpose grease, 500 g	SELFFIX HOME & DIY	13.80	1	13.80
Subtotal, actuator and sensor repair					**102.89**
Grand total (total amount spent)					**692.71**

Table 7: Project expenditure